

Developing Games as "Translation Devices" for Quantum Annealing and Validating its Social Implementation

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Abstract

This research positions interactive games as "translation devices" to facilitate the social implementation of quantum computing. We implemented a hybrid framework connecting a Unity-based game engine with a backend quantum annealing machine, transforming quantum optimization outputs into immediate evasive body motions during gameplay.

Unlike existing quantum games that primarily use quantum systems as sources of randomness, our system deploys quantum annealing as a goal-oriented computational engine for real-time NPC posture optimization. The resulting emergent movements—often fragmented, acrobatic, and comical—visually externalize the otherwise invisible behavior of constrained optimization under discrete state spaces.

This approach explores how embodied interaction and gamification can foster intuitive understanding of quantum computation while introducing new forms of strategic gameplay driven by optimization-based behaviors.

CCS Concepts

• **Applied computing** → **Computer games; Interactive learning environments**; • **Computer systems organization** → **Quantum computing; Heterogeneous (hybrid) systems**; • **Human-centered computing** → **Visualization**.

Keywords

Quantum Computing, Quantum Annealing, Game Design, Human-Computer Interaction, Combinatorial Optimization

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1 Introduction

Quantum computing has attracted significant attention as a next-generation computational paradigm capable of solving problems beyond classical architectures [Skult et al. 2024]. However, quantum mechanics remains highly abstract and counter-intuitive to everyday experience, creating substantial barriers to public understanding and the development of "quantum literacy" [Nita et al. 2021; Piispanen et al. 2025].

To bridge this gap, researchers have explored interactive games as a means of fostering intuitive understanding through embodied interaction [Piispanen et al. 2025]. While the ecosystem of "quantum games" continues to expand, the integration of actual quantum hardware into gameplay remains limited. Most existing implementations utilize quantum backends primarily as sources of randomness or procedural variation rather than as goal-oriented computational systems that actively influence gameplay [Skult et al. 2024]. Consequently, games that employ quantum computation as a core optimization engine for real-time decision-making and character control remain rare.

This research addresses this gap through *AFTER IMAGE*, an interactive game that embeds a quantum annealing machine directly into the gameplay loop. Rather than generating stochastic events, the quantum computer performs real-time posture optimization to drive NPC evasive behavior. Through this design, the invisible process of constrained optimization becomes perceptible as physical motion and strategic interaction, positioning games as "translation devices" between quantum computation and human experience.

2 AFTER IMAGE: Game Design and Mechanics

AFTER IMAGE is a competitive shooting game in which a human player directly confronts a quantum annealing machine embodied as an enemy NPC.



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Figure 1: The visual concept of *AFTER IMAGE*.

During the aiming phase, the player places multiple firing trajectories to surround the target and reduce its feasible escape space. Upon firing, the generated trajectory data is transmitted to a backend quantum annealing machine, which searches a large combinatorial solution space for an evasive posture that minimizes incoming damage.

As illustrated in Figure 2, the player attempts to strategically corner the target, while the NPC responds with quantum-optimized evasive maneuvers. Because the optimization operates on a discrete and constrained state space, the resulting movements often appear fragmented, acrobatic, and occasionally comical, visually exposing the behavior of optimization-driven computation.

The game adopts a three-round turn-based structure. The number of projectiles increases from 3 to 5 and then 8 bullets across successive rounds, forcing both the player and the optimization system to operate under increasingly constrained conditions. Damage scores are calculated according to the body parts struck, and the final outcome is determined by the cumulative score across all rounds.

Through this adversarial structure, combinatorial optimization becomes a tangible and strategic gameplay experience.

3 Quantum Posture Optimization Logic

To achieve real-time responsive evasion, *AFTER IMAGE* leverages a backend quantum annealing machine [Kadowaki and Nishimori 1998] by formulating the character’s kinematics into a Quadratic Unconstrained Binary Optimization (QUBO) expression. Specifically, the frontend game loop is implemented in Unity, while the QUBO model is formulated and executed using the Fixstars Amplify platform.

The optimization objective is to determine the NPC’s joint configuration to minimize cumulative bullet damage. The controllable degrees of freedom (DoFs) comprise the overall global yaw orientation (discretized into 5-degree increments, 72 states) and ten primary joint angles (ankle, knee, hip, pelvis, lumbar/thoracic/cervical vertebrae, shoulder, elbow, and wrist). Simple joints feature 3–10 discrete states, while high-DoF joints (e.g., shoulder and hip joints) feature ≈ 250 patterns. Simultaneously solving this entire structure yields an intractable state space of $\approx 2 \times 10^{18}$ combinations. To bypass this real-time computational bottleneck, we decompose the execution pipeline into five sequential sub-models: (1) a Global Yaw



(a) Aiming phase with multi-angle trajectories.



(b) Quantum-optimized evasion phase.

Figure 2: The core gameplay loop of *AFTER IMAGE*: (a) The player sets up multiple trajectories to trap the target. (b) The backend quantum annealing machine calculates the optimal physical posture for the NPC to minimize damage.

Model, (2) an Axial Backbone Model, and (3) three independent Limb Models (right leg, right arm, left arm).

Mathematically, let $x_{j,p} \in \{0, 1\}$ be a binary variable indicating if joint j selects posture pattern p . The strict one-hot configuration constraint and the total damage Hamiltonian are formulated as follows:

$$H_{\text{constraint}} = \lambda \sum_j \left(1 - \sum_p x_{j,p} \right)^2 \quad (1)$$

$$H_{\text{damage}} = \sum_b D_b \left(\sum_j \sum_p H_{j,p,b} \cdot x_{j,p} \right) \quad (2)$$

where λ is a penalty coefficient, D_b represents the damage weight of bullet b , and $H_{j,p,b} \in \{0, 1\}$ is a pre-computed bullet-hit tensor mapping segment-line intersections. The total minimized cost is $H_{\text{total}} = H_{\text{damage}} + H_{\text{constraint}}$.

To track the kinematic cascade within the Limb Models, consider the left arm as a representative example. Since the ancestral Axial Backbone Model pre-determines the shoulder angle as a constant, the remaining variables for the elbow (e_i), wrist (w_j), and hand (h_k) dictate the localized segment damage:

$$H_{\text{damage}}^{\text{left_arm}} = H_{\text{upper_arm}}^{(1)} + H_{\text{fore_arm}}^{(2)} + H_{\text{palm}}^{(3)} \quad (3)$$

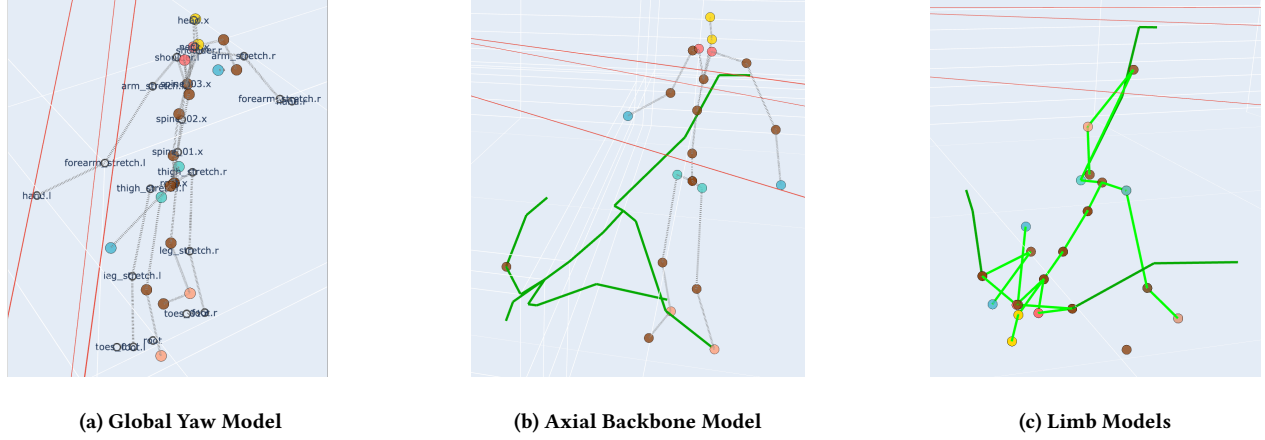


Figure 3: Sequential posture resolution of the NPC through the decomposed sub-models.

$$H_{\text{upper_arm}}^{(1)} = \sum_b D_b \sum_i H_{\text{upper},i,b} \cdot e_i \quad (1\text{st-order}) \quad (4)$$

$$H_{\text{fore_arm}}^{(2)} = \sum_b D_b \sum_i \sum_j H_{\text{fore},i,j,b} \cdot e_i w_j \quad (2\text{nd-order}) \quad (5)$$

$$H_{\text{palm}}^{(3)} = \sum_b D_b \sum_i \sum_j \sum_k H_{\text{palm},i,j,k,b} \cdot e_i w_j h_k \quad (3\text{rd-order}) \quad (6)$$

To ensure compatibility with the quadratic QUBO formulation required by the annealing hardware, higher-order interaction terms are reduced using standard quadratization techniques.

4 Evaluation and Future Work

To evaluate real-time stability and player reception, *AFTER IMAGE* was live-demonstrated at the Taipei Game Show 2026 (Figure 4). Approximately 200 attendees actively played the game, providing positive qualitative feedback. Players highly praised the emergent behaviors of the enemy NPC; despite assuming highly discretized and often comical physical postures due to state sampling, the NPC consistently and accurately evaded incoming bullet trajectories. This juxtaposition of a humorous aesthetic and rigorous evasion validated that our proposed QUBO pipeline can govern real-time, goal-oriented kinematics under demanding exhibition conditions.

Our primary future trajectory focuses on deepening quantum literacy by gamifying the inherent computational limitations of the QUBO framework. Specifically, we aim to map known limitations—such as the tendency to violate hard constraints due to the penalty-method formulation and the difficulty of handling sequential decision-making under Markov Decision Processes (MDPs)—directly onto player-facing strategic mechanics. By transforming theoretical hardware bottlenecks into core gameplay challenges, we aim to reinforce an educational pipeline where optimizing gameplay instinctively fosters a realistic understanding of quantum hardware boundaries among the general public.

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Figure 4: The physical exhibition booth of *AFTER IMAGE* at the Taipei Game Show 2026.

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